

deliver a multiple of the mystical Pi, with possibly infinite precision. Clearly, this is not an efficient use of a computer as Pi can be computed analytically, that is, in a mathematical form, but the method can give some users more intuition about the subject matter than lines of equations. Some people's brains and intuitions are oriented in such a way that they are more capable of getting a point in such a manner (I count myself one of those). The computer might not be natural to our human brain; neither is mathematics.

I am not a "native" mathematician, that is, I am someone who does not speak mathematics as a native language, but someone who speaks it with a trace of a foreign accent. For I am not interested in mathematical properties *per se*, only in the application, while a mathematician would be interested in improving mathematics (via theorems and proofs). I proved incapable of concentrating on deciphering a single equation unless I was motivated by a real problem (with a modicum of greed); thus most of what I know comes from derivatives trading – options pushed me to study the math of probability. Many compulsive gamblers, who otherwise would be of middling intelligence, acquire remarkable card-counting skills thanks to their passionate greed.

Another analogy would be with grammar; mathematics is often tedious and insightful grammar. There are those who are interested in grammar for grammar's sake, and those interested in avoiding solecisms while writing documents. We are called "quants" – like physicists, we have more interest in the employment of the mathematical tool than in the tool itself. Mathematicians are born, never made. Physicists and quants too. I do not care about the "elegance" and "quality" of the mathematics I use so long as I can get the point right. I have recourse to Monte Carlo machines whenever I can. They can get the work done. They are also far more pedagogical, and I will use them in this book for the examples.

Indeed, probability is an introspective field of inquiry, as it affects more than one science, particularly the mother of all sciences; that of knowledge. It is impossible to assess the quality of the knowledge we are gathering without allowing a share of randomness in the manner it is obtained and cleaning the argument from the chance coincidence that could have seeped into its construction. In science, probability and information are treated in exactly the same manner. Literally every

great thinker has dabbled with it, most of them obsessively. The two greatest minds to me, Einstein and Keynes, both started their intellectual journeys with it. Einstein wrote a major paper in 1905, in which he was almost the first to examine in probabilistic terms the succession of random events, namely the evolution of suspended particles in a stationary liquid. His theory on the theory of the Brownian movement can be used as the backbone of the random walk theories used in financial modeling. As for Keynes, to the literate person he is not the political economist that tweed-clad leftists love to quote, but the author of the magisterial, introspective, and potent *Treatise on Probability*. For before his venturing into the murky field of political economy, Keynes was a probabilist. He also had other interesting attributes (he blew up trading his account after experiencing excessive opulence – people's understanding of probability does not translate into their behavior).

The reader can guess that the next step from such probabilistic introspection is to get drawn into philosophy, particularly the branch of philosophy that concerns itself with knowledge, called epistemology or methodology, or philosophy of science, popularized by such persons as Karl Popper and George Soros. We will not get into the topic until later in the book.

Fun In My Attic

MAKING HISTORY

In the early 1990s, like many of my friends in quantitative finance, I became addicted to the various Monte Carlo engines, which I taught myself to build, thrilled to feel that I was generating history, a *Demiurgus*. It can be electrifying to generate virtual histories and watch the dispersion between the various results. Such dispersion is indicative of the degree of resistance to randomness. This is where I am convinced that I have been extremely lucky in my choice of career: one of the attractive aspects of my profession as a quantitative option trader is that